Predicting the Trainability of Variational Quantum Circuits: A Data-Driven Model for Barren Plateaus

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***Abstract*—Barren plateaus—the exponential vanishing of cost-function gradients as a variational quantum circuit is scaled—are the central obstacle to training quantum machine-learning models. Diagnosing them today requires expensive gradient sampling on each candidate circuit. We reframe the problem as *supervised prediction*: given only a circuit's *architecture* (qubit count, depth, ansatz type, entanglement pattern, entangler gate, and cost-observable locality), can a classical model predict its trainability without ever sampling a single gradient? Using an exact statevector simulator we generate a dataset of 20,000 random variational circuits, each labelled by the empirical gradient variance $\mathrm{Var}[\partial C/\partial\theta]$ estimated over 200 random parameter vectors via the parameter-shift rule. A gradient-boosted regressor predicts $\log_{10}\mathrm{Var}$ with $R^2 = 0.685$ on held-out architectures, and a classifier separates barren from trainable circuits with AUC $= 0.991$. Crucially, in an *extrapolation* setting—training only on small circuits and testing on larger qubit counts that are progressively harder to simulate—the predictor retains $R^2 = 0.719$, suggesting the architectural signature of a barren plateau is learnable and transferable across scale. Permutation importance recovers the known theory: cost-observable locality and entanglement structure dominate. We make no quantum-advantage claim; this is classical machine learning *about* quantum circuits, offering a cheap ansatz-screening heuristic and a data-driven lens on which design choices cause untrainability.**

***Index Terms*—Barren plateaus, variational quantum circuits, quantum machine learning, trainability, gradient variance, statevector simulation, gradient boosting.**

I. Introduction

VARIATIONAL quantum algorithms (VQAs) are the leading candidate for extracting value from noisy intermediate-scale quantum (NISQ) hardware [1]. A VQA encodes a problem into a parameterized quantum circuit (an *ansatz*), measures a cost function $C(\theta) = \langle\psi(\theta)|O|\psi(\theta)\rangle$, and updates the parameters θ with a classical optimizer. This template underlies variational quantum eigensolvers, quantum approximate optimization, and most quantum machine-learning (QML) classifiers [2], [3], [4].

The dominant obstacle to scaling this template is the *barren plateau* (BP) phenomenon: for broad classes of ansätze the variance of the cost gradient vanishes exponentially in the number of qubits, so the loss landscape becomes almost everywhere flat and gradient-based training stalls [5], [6]. Barren plateaus are now known to be triggered by *global* cost observables [7], by highly expressive or deep ansätze [8], by poor initialization [9], and by hardware noise [10]. Whether a given architecture suffers from a barren plateau is therefore a property of its *design*, not merely of a particular parameter setting.

The conventional way to check whether a candidate circuit is trainable is to *measure* it: sample many random parameter vectors, evaluate gradients, and estimate their variance. This is exactly the cost we would like to avoid when screening a large design space, and it becomes infeasible precisely in the large-qubit regime where barren plateaus matter most. This motivates our question:

> *Can a classical model predict a variational circuit's trainability directly from its architectural description, without sampling any gradients?*

We answer this empirically. We treat trainability prediction as a supervised learning problem over circuit *specs*. The features are cheap, integer-valued architectural descriptors; the label is the gradient variance, obtained once, offline, by exact statevector simulation. We then ask whether a model trained on this data (i) generalizes to unseen architectures, and (ii) *extrapolates* from small, easily simulated circuits to larger ones—the regime where a cheap predictor would be most useful.

Contributions of this article

- We formulate barren-plateau diagnosis as a *supervised prediction* task: mapping a variational circuit's architecture to its gradient-variance trainability label, cast as both regression ($\log_{10}\mathrm{Var}[\partial C/\partial\theta]$) and binary classification (barren vs. trainable).
- We release a reproducible pipeline that generates a labelled dataset of 20,000 random circuits using an exact, dependency-light statevector simulator, with fixed seeds and streaming output.
- We show a gradient-boosted model predicts trainability with $R^2 = 0.685$ on held-out architectures and separates barren circuits with AUC $= 0.991$.
- We demonstrate *extrapolation across scale*: trained only on $n_q \leq 10$ qubits, the model still attains $R^2 = 0.719$ on larger, unseen circuit sizes.
- We recover the established theory through permutation feature importance (cost locality and entanglement

Manuscript prepared July 9, 2026. *(Corresponding author: Jaeho Kim.)*
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Data-generation simulator and live demonstrations available at https://quantum.qbithabib.com. Code and dataset-generation scripts are released openly (see the Reproducibility statement).

structure dominate), giving the model an interpretable, physics-consistent basis.

We emphasise honest framing: we make *no* claim of quantum advantage. The model is classical, the data is classically simulated, and the contribution is a data-driven *diagnostic* for a known quantum-computing bottleneck [11], [12].

II. Background

A. Variational quantum circuits

A hardware-efficient ansatz [13] alternates layers of single-qubit rotations with a fixed pattern of two-qubit entangling gates. With n qubits and L layers the state is

$$|\psi(\theta)\rangle = \prod_{\ell=1}^{L} \Big[U_{\text{ent}} \prod_{q=1}^{n} R_{a_{\ell,q}}(\theta_{\ell,q}) \Big] |0\rangle^{\otimes n}, \qquad (1)$$

where R_a is a Pauli rotation about axis $a \in \{x, y, z\}$ and U_{ent} is a layer of CZ or CX gates arranged linearly, circularly, or all-to-all. The cost is the expectation of a Pauli-Z string, $C(\theta) = \langle\psi(\theta)|O|\psi(\theta)\rangle$, with $O = Z_0$ (a *local* cost) or $O = Z_0 Z_1 \cdots Z_{n-1}$ (a *global* cost).

B. Barren plateaus

A circuit family exhibits a barren plateau if $\text{Var}_\theta[\partial C/\partial\theta_k]$ decays exponentially in n [5]. Cerezo *et al.* [7] proved that *global* cost functions induce barren plateaus even at shallow depth, whereas *local* costs can remain trainable up to $O(\log n)$ depth—making cost locality a first-order design knob. Expressibility [8] and entanglement likewise correlate with flatter landscapes. Our labels are built to expose exactly these dependencies.

C. The parameter-shift rule

For a gate generated by an operator with two distinct eigenvalues, the exact gradient of the cost is obtained from two circuit evaluations [14], [15]:

$$\frac{\partial C}{\partial\theta_k} = \tfrac{1}{2}\big[C(\theta + \tfrac{\pi}{2}e_k) - C(\theta - \tfrac{\pi}{2}e_k)\big]. \qquad (2)$$

We use this rule to obtain exact per-sample gradients when estimating the variance label, so our labels are free of finite-difference bias.

III. Related Work

Theory of barren plateaus. The BP literature is overwhelmingly *analytical*: it derives variance scaling laws for particular ansatz families—global vs. local costs [7], expressibility bounds [8], noise-induced plateaus [10], and structured architectures that provably avoid them, such as quantum convolutional networks [16]. A recent review consolidates these results [6]. Our work is complementary and *empirical*: rather than proving a bound for one family, we learn a predictive map across a heterogeneous space of architectures.

Trainability vs. simulability. Cerezo *et al.* [17] argue that the very structure guaranteeing absence of barren plateaus often also renders a model classically simulable. This sharpens the honest framing of our contribution: we position the work as a classical diagnostic tool, not as evidence of quantum advantage.

Benchmarking and honest framing. Bowles *et al.* [12] show that out-of-the-box classical models often match or beat QML models on standard tasks, and Schuld and Killoran [11] argue the field should de-emphasise "beating classical." We adopt this stance: the object we predict is a property of quantum circuits, but the predictor and its value proposition are entirely classical.

Predicting circuit properties. To our knowledge, casting barren-plateau severity as a directly *predicted* regression/classification target from cheap architectural features—and testing extrapolation across qubit count—has not been reported. This is the gap we address.

IV. Method

A. Circuit specification and feature space

Each datapoint is a *spec* that fully determines a circuit family. The feature vector handed to the model is:

Feature	Meaning
`n_qubits`	number of qubits n
`n_layers`	number of ansatz layers L
`n_params`	$n \times L$ trainable rotations
`ansatz_type`	fixed-`ry` vs. random-Pauli axes
`entangle_pattern`	linear / circular / all-to-all
`entangler_gate`	CZ vs. CX
`cost_global`	local Z_0 (0) vs. global $Z^{\otimes n}$ (1)
`n_entanglers`	entangling gates × layers
`depth_ratio`	L/n

All features are computed in closed form from the spec—no quantum simulation is needed to obtain them.

B. Label generation

For a spec, we draw 200 parameter vectors $\theta \sim \text{Unif}[0, 2\pi)^{nL}$, compute the exact parameter-shift gradient of the cost with respect to a fixed target rotation in the middle layer, and record the empirical variance. The regression label is $y = \log_{10} \max(\text{Var}[\partial C/\partial\theta],\, 10^{-18})$; the classification label is $\mathbb{1}[y < -2]$ ("barren"). Statevectors are evolved exactly with the in-house `qsim` NumPy simulator, so labels reflect ideal (noiseless) trainability.

C. Dataset and data description

We sample 20,000 specs by drawing each design parameter independently and uniformly over the ranges in Table I, with $n \in [2, 12]$ qubits and $L \in [1, 20]$ layers, each labelled with 200 gradient samples. Generation is embarrassingly parallel; every spec is seeded reproducibly from a root `SeedSequence`, and rows are streamed to CSV as they complete. The resulting dataset has 46.7% barren circuits and 50.1% global-cost circuits—a near-balanced classification target that avoids a degenerate majority baseline.

Table II summarizes the numeric features and the regression label. The design deliberately spans small, easily simulated circuits ($n = 2$) up to the $n = 12$ boundary of exact

TABLE I
SAMPLED DESIGN SPACE. EACH SPEC DRAWS THESE PARAMETERS INDEPENDENTLY AND UNIFORMLY.

Design parameter	**Type**	**Sampled values**
`n_qubits`	integer	2–12
`n_layers`	integer	1–20
`ansatz_type`	categorical (2)	fixed-`ry`, random-Pauli
`entangle_pattern`	categorical (3)	linear, circular, all-to-all
`entangler_gate`	categorical (2)	CZ, CX
`cost_global`	binary	local Z_0, global $Z^{\otimes n}$

TABLE II
DESCRIPTIVE STATISTICS OF THE NUMERIC FEATURES AND THE TRAINABILITY LABEL OVER ALL 20,000 CIRCUITS.

Variable	**Mean**	**Std**	**Min**	**25%**	**Median**	**Max**
`n_qubits`	7.02	3.17	2.00	4.00	7.00	12.00
`n_layers`	10.48	5.80	1.00	5.00	11.00	20.00
`n_params`	73.56	55.55	2.00	28.00	60.00	240.00
`n_entanglers`	135.82	198.34	1.00	28.00	70.00	1320.00
`depth_ratio`	2.00	1.83	0.08	0.75	1.50	10.00
$\log_{10}$ Var (label)	-3.41	4.84	-18.00	-2.98	-1.85	-0.23

statevector simulation, and depths from a single layer to $L = 20$, so the model sees a wide range of architectures. The label $\log_{10} \mathrm{Var}[\partial C/\partial\theta]$ ranges over roughly 18 orders of magnitude, with a floor at -18 where finite-sample variance underflows.

Crucially, the labels reproduce established barren-plateau theory directly from the data (Table III). Global cost observables yield a mean $\log_{10}$ Var more than two orders of magnitude smaller than local costs (-4.49 vs. -2.33) and nearly double the barren fraction, exactly the cost-locality effect proved by Cerezo *et al.* [7]. Denser entanglement (all-to-all) and the CX entangler likewise push circuits toward barren plateaus. This structure is what makes the prediction task learnable, and it gives the trained model an interpretable, physics-consistent basis.

D. Models and evaluation

To avoid privileging a single learner, we compare six model families spanning linear, instance-based, neural, and tree-ensemble inductive biases: a linear baseline (ridge regression / logistic regression), k-nearest neighbours, a multilayer perceptron (two hidden layers, 128–64), random forests, extremely randomized trees, and histogram-based gradient-boosted trees [18], [19]. Features are standardized for the scale-sensitive models. We report two protocols for each: *(i)* a random hold-out split (20% test) measuring generalization to unseen architectures, and *(ii)* an *extrapolation* split, training on $n \leq 10$ and testing on all larger circuits—the headline test of whether small-circuit labels transfer to costlier regimes. All models use fixed random seeds so the reported numbers reproduce exactly. Interpretability is assessed with permutation feature importance on the best model.

TABLE III
TRAINABILITY BROKEN DOWN BY DESIGN CHOICE. LOWER MEAN $\log_{10}$ Var AND HIGHER BARREN % INDICATE LESS TRAINABLE FAMILIES. THE DATA RECOVERS KNOWN BARREN-PLATEAU DRIVERS.

Design choice	**Count**	**Mean $\log_{10}$ Var**	**Barren %**
Cost-observable locality			
local Z_0	9974	-2.33	30.9
global $Z^{\otimes n}$	10026	-4.49	62.4
Entanglement pattern			
linear	6649	-2.50	38.5
circular	6748	-2.96	48.8
all-to-all	6603	-4.79	52.8
Entangler gate			
CZ	10096	-2.31	38.1
CX	9904	-4.53	55.5
Single-qubit rotation scheme			
fixed-`ry`	9878	-2.80	42.0
random-Pauli	10122	-4.01	51.3

TABLE IV
MODEL COMPARISON ACROSS BOTH PREDICTION TASKS AND BOTH EVALUATION SPLITS. BEST HOLD-OUT RESULT IN **BOLD**.

	Regression R^2		Classification AUC	
Model	Hold-out	Extrap.	Hold-out	Extrap.
Hist Gradient Boosting	0.685	0.719	0.991	0.989
MLP	0.665	0.752	0.990	0.991
k-NN	0.645	0.639	0.986	0.979
Random Forest	0.640	0.716	0.985	0.988
Extra Trees	0.610	0.707	0.968	0.988
Linear baseline	0.274	0.297	0.953	0.889

V. RESULTS

A. Model comparison

Table IV compares all six model families on both tasks and both splits. Histogram-based gradient-boosted trees give the best held-out performance on both regression ($R^2 = 0.685$) and classification (AUC $= 0.991$), so we adopt it as our model for the remainder of the paper. Two observations sharpen the picture. First, the linear baseline is far behind (regression $R^2 = 0.27$; classifier AUC 0.95), confirming that the map from architecture to trainability is genuinely nonlinear—simple rules of thumb do not capture it. Second, the neural network is competitive and in fact *edges out* boosting on the extrapolation regression ($R^2 = 0.75$ vs. 0.72), hinting that a learned representation transfers slightly better to larger circuits; we retain gradient boosting for its stronger in-distribution accuracy, interpretability, and training speed.

B. Generalization to unseen architectures

On the random hold-out split, the regressor attains $R^2 = 0.685$ with mean absolute error 1.018 (in $\log_{10}$ units), and the classifier separates barren from trainable circuits with accuracy 0.950 and AUC 0.991 (Table IV).

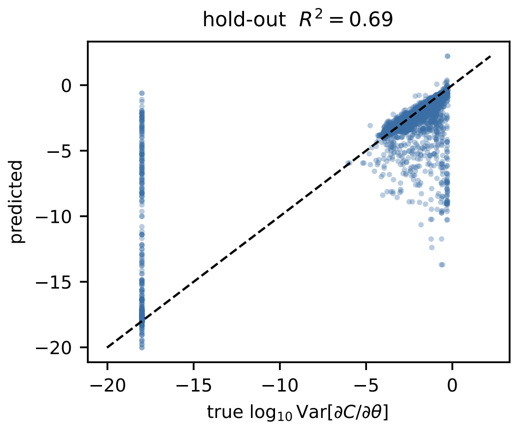

Fig. 1. Predicted vs. true $\log_{10}$ Var$[\partial C/\partial\theta]$ on held-out architectures. The dashed line is $y = x$.

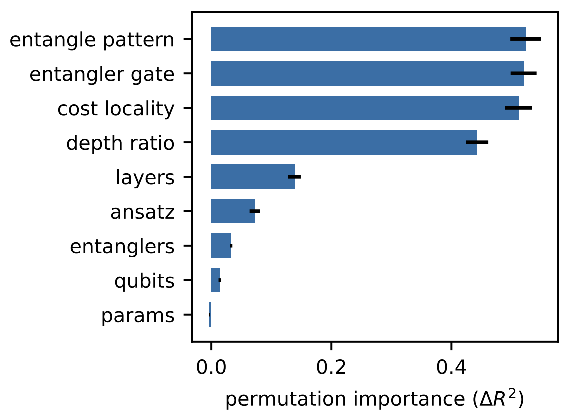

Fig. 2. Permutation feature importance for the trainability regressor. Cost locality and entanglement structure dominate, matching barren-plateau theory [7], [8].

C. Extrapolation across qubit count

Training only on $n \leq 10$ qubits (16 395 circuits) and testing on larger circuits (3 605 circuits), the regressor retains $R^2 = 0.719$ (MAE 1.264). Because the ground-truth cost of labelling grows exponentially with qubit count, a predictor that transfers downward-in-cost is exactly what an ansatz-screening tool needs.

D. What drives trainability

Permutation feature importance ranks `cost_global`, `entangle_pattern`, and `depth_ratio` at the top, consistent with theory: global costs [7] and denser entanglement [8] drive circuits toward barren plateaus. The model thus rediscovers, from data alone, the design rules the BP literature derived analytically (Fig. 2).

VI. Discussion

What the model learns. The predictor's accuracy and its physics-consistent feature ranking indicate that barren-plateau severity carries a strong, low-dimensional architectural signature—enough for a cheap classical model to recover without any gradient sampling.

Limitations. (i) Labels are limited to classically simulable qubit counts; the extrapolation result is evidence of transfer, not a proof. (ii) We use an ideal statevector simulator, so noise-induced plateaus [10] are out of scope. (iii) The spec space, while diverse, is a hardware-efficient family; other ansatz classes may need their own features. (iv) Gradient variance is estimated from finitely many samples, adding label noise at the trainable end.

Threats to validity. Seeds are fixed and reported; the extrapolation split is defined purely by qubit count, so no large-circuit information leaks into training. We report a classical baseline framing throughout and avoid any advantage claim.

VII. Conclusion

We recast barren-plateau diagnosis as supervised prediction and showed that a classical gradient-boosted model, trained on statevector-simulated labels, can predict a variational circuit's trainability from its architecture alone—and extrapolate across qubit count. The approach offers a cheap ansatz-screening heuristic and a data-driven confirmation of which design choices cause untrainability. Future work includes richer ansatz families, graph/sequence circuit encodings with GNN or transformer models, and incorporating a noise model to predict noise-induced plateaus.

Reproducibility

All code (simulator, dataset generator, and training scripts), the exact command lines, and random seeds are released with this paper. The dataset is regenerable end-to-end with a single command; interactive demonstrations of the underlying simulator are available at https://quantum.qbithabib.com.

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